



Acetaldehyde oxidation at low and intermediate temperatures: An experimental and kinetic modeling investigation

Xiaoyuan Zhang^{a,b}, Lili Ye^b, Yuyang Li^{a,b,*}, Yan Zhang^c, Chuangchuang Cao^c,
Jiuzhong Yang^c, Zhongyue Zhou^b, Zhen Huang^b, Fei Qi^{a,b,*}

^a Collaborative Innovation Center for Advanced Ship and Deep-Sea Exploration (CISSE), Shanghai 200240, PR China

^b Key Laboratory for Power Machinery and Engineering of MOE, Shanghai Jiao Tong University, Shanghai 200240, PR China

^c National Synchrotron Radiation Laboratory, University of Science and Technology of China, Hefei, Anhui 230029, PR China

ARTICLE INFO

Article history:

Received 3 October 2017

Revised 20 November 2017

Accepted 18 January 2018

Keywords:

Acetaldehyde

Low-temperature oxidation

SVUV-PIMS

Kinetic model

Low-temperature methyl oxidation mechanism

ABSTRACT

Acetaldehyde oxidation was investigated in a jet-stirred reactor at temperatures from 460 to 900 K, equivalence ratios from 0.5 to 4.0 and pressures of 710–720 Torr. Reactive intermediates under low-temperature conditions, such as methylperoxy, methylhydroperoxide and ketohydroperoxide were detected using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). A kinetic model for acetaldehyde oxidation was constructed by incorporating recent theoretical and modeling progress of acetaldehyde kinetics, as well as the calculated results of H-atom abstraction reactions of acetaldehyde by acetylperoxy, methylperoxy and methoxy in this work. The present model was then comprehensively validated against the measurements in this work and the experimental data from literature. Modeling analysis reveals that the main chain-branching pathways of acetaldehyde under low-temperature oxidation conditions are the decomposition of acetylhydroperoxide and methylhydroperoxide. The second O₂-addition process was verified to exist in the low-temperature oxidation of acetaldehyde as ketohydroperoxide was observed, while its contribution to the overall reactivity was found to be minor under present investigated conditions. On the other hand, the sub-mechanism of acetylperoxy is crucial in the low-temperature regime while the methyl oxidation mechanism dominates in the negative temperature coefficient (NTC) region at all equivalence ratios investigated in this work.

© 2018 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

1. Introduction

Acetaldehyde (CH₃CHO) is one of the toxic pollutants that can cause allergies, liver diseases and carcinogen risks [1]. It can be produced from combustion process [2,3] since it is a typical combustion intermediate/product of fossil fuels and biofuels (especially alcohols). Understanding the combustion chemistry of acetaldehyde is not only needed for emission reduction, but also critical in the development of kinetic models for hydrocarbon fuels and biofuels. Besides, the extraordinarily weak C–C bond in acetyl (CH₃CO) (~10 kcal/mol [4]) makes it readily decompose to methyl (CH₃) and CO, implying a strong relationship between the acetaldehyde chemistry and the methyl chemistry.

The previous studies of the high-temperature chemistry of acetaldehyde before 2015 were reviewed in detail by Christensen et al. [5], and thus only recent progress is briefly summarized herein. Sivaramakrishnan et al. [6] performed theoretical calculations of the C₂H₄O and C₂H₅O potential energy surfaces (PES) and updated the pyrolysis model of acetaldehyde. Mével et al. [7] measured ignition delay times of acetaldehyde in a shock tube at 1295–1580 K and 306–404 kPa. Tao et al. [8] investigated the laminar premixed flames of acetaldehyde and identified about 40 flame species. Christensen et al. [5] measured the laminar burning velocities of acetaldehyde/air mixture at atmospheric pressure and various initial temperatures. Furthermore, Christensen and Konnov [9] recently measured the laminar burning velocities of diacetyl and updated the sub-mechanisms of acetaldehyde and CH₃CO in their model.

The low-temperature combustion studies of acetaldehyde can date back to 1882 when its beautiful bluish flame (so-called cool flame) was observed by Perkin [10]. However, there are still few experimental investigations on the low-temperature oxidation of acetaldehyde [11] to date. Gray et al. [12] investigated the

* Corresponding authors at: Key Laboratory for Power Machinery and Engineering of MOE, Shanghai Jiao Tong University, Shanghai 200240, PR China.

E-mail addresses: yuygli@sjtu.edu.cn, yuygli@ustc.edu.cn (Y. Li), fqi@sjtu.edu.cn (F. Qi).

ignition and cool flame behaviors of acetaldehyde using a continuous well stirred flow reactor. However, large uncertainties existed in the determination of the heat exchange coefficient, which caused a big influence on the simulation results [11,13]. Later, Kaiser et al. [14] investigated the acetaldehyde oxidation in a low-pressure static reactor at 553 and 713 K using molecular-beam mass spectrometry and gas chromatography. Besides the major species like CH_3CHO , CO and CO_2 , they observed the mass peaks of $m/z = 48$ and 34 and directly attributed them to methyl hydroperoxide (CH_3OOH) and hydroperoxide (H_2O_2). It is recognized that mass spectrometric signals are susceptible to the isomeric structures, the ^{13}C and ^{18}O isotopomers of smaller species and the fragments of larger species. Therefore, more experimental evidences are needed to confirm the existence of these peroxides. Besides, because of the very rapid initial reaction rate, the concentrations of the products could not be reliably measured until approximately 50% of acetaldehyde was consumed and the initial consumption rate of acetaldehyde was uncertain. Recently, Barari et al. [15,16] qualitatively measured the intermediates in the acetaldehyde oxidation at 500–700 K using multiplexed photoionization mass spectrometry with synchrotron radiation. The chlorine atom produced by laser-pulsed photolysis can react with acetaldehyde to form the parent radicals. They observed ketene (CH_2CO) and methylperoxy (CH_3OO), but could not quantify all the measured species.

For the model development, Halstead et al. [17] proposed a 14-step mathematical model for acetaldehyde to interpret its cool-flame phenomenon. They considered acetyl played an exclusive role in the chain-branching process through $\text{CH}_3\text{CO} \rightarrow \text{CH}_3\text{C}(\text{O})\text{OO} \rightarrow \text{CH}_3\text{C}(\text{O})\text{OOH} \rightarrow \text{CH}_3\text{C}(\text{O})\text{O} + \text{OH}$, which was supported by the theoretical work of Felton et al. [18] and the detailed kinetic modeling work of Cavanagh et al. [19]. However, Gibson et al. [20] concluded the cool flame phenomena of acetaldehyde originated from methyl oxidation and the degenerate branching via CH_3OOH ($\text{CH}_3 \rightarrow \text{CH}_3\text{OO} \rightarrow \text{CH}_3\text{OOH} \rightarrow \text{CH}_3\text{O} + \text{OH}$). In the studies of Kaiser et al. [14], they concluded the competition between the radical decomposition reaction ($\text{CH}_3\text{CO} = \text{CH}_3 + \text{CO}$) and the addition of O_2 to acetyl ($\text{CH}_3\text{CO} + \text{O}_2 = \text{CH}_3\text{C}(\text{O})\text{OO}$), is the key factor determining the dominant chain-branching pathway. Very recently, Pelucchi et al. [11,13] developed the low-temperature oxidation model for acetaldehyde as well as C3–C4 aldehydes and considered both aforementioned chain-branching pathways and validated their model against the previous experiment in a continuous well stirred flow reactor [12]. In summary, there are still inadequate investigations on the chemical structure measurements for the low-temperature oxidation of acetaldehyde, especially the unambiguous identification and quantification of some reactive key species. Besides, the chain-branching reactions which determine the low-temperature oxidation rate of acetaldehyde are still in debate.

This work has three main targets. Firstly, the species concentration information was provided for the acetaldehyde oxidation in a jet-stirred reactor (JSR) at temperatures from 460 to 900 K, equivalence ratios from 0.5 to 4.0 and 710–720 Torr. A variety of intermediates, especially reactive species, were identified and quantified by the synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Secondly, critical reactions in the low-temperature acetaldehyde oxidation were investigated, combining with the quantum chemical calculations for the H-atom abstraction reactions of acetaldehyde by $\text{CH}_3\text{C}(\text{O})\text{OO}$, CH_3OO and CH_3O due to the limited investigations in literature. Thirdly, a kinetic model of acetaldehyde was developed and validated against the present experimental data and those from literature over the temperatures of 400–2100 K, pressures of 0.02–10 atm and equivalence ratios between 0.09 and pyrolysis. Particularly, the chain-branching reactions of acetaldehyde under low-temperature conditions were discussed.

Table 1

Experimental conditions in the present work.

No.	ϕ	P (Torr)	Fuel%	$\text{O}_2\%$	Ar%
1	0.5	710	2	10	88
2	1.0	720	2	5	93
3	4.0	720	2	1.25	96.75

2. Experimental method

The experiments were conducted at the BL03U beamline of the National Synchrotron Radiation Laboratory. Introduction of this VUV undulator beamline can be found elsewhere [21]. The JSR used in this work has a similar design to our previous reactor [22] complying with the rules established by Matras and Villermaux [23]. The volume of the JSR is 100 cm^3 and a quartz sampling nozzle with a ~ 70 μm orifice is welded into the sphere. Quartz wall effects are considered as negligible based on previous research [24]. A short annular preheating zone is attached to the reactor for preheating the inlet mixtures in a very short period of time (a few percent of the residence time) to achieve more reliable temperature homogeneity in the reactor [25]. Both the reactor and the preheating zone were heated by the heating jackets and controlled by the temperature-controlling units. The reaction temperature was measured by a K-type thermocouple located in the center of the reactor, which had a fluctuation within 2 K for each setting temperature during the experiments. The schematic diagram of the JSR can be found in the *Supplementary Materials*. No complex patterns, like oscillatory cool flames or multi-stage ignitions [20], were observed in this work.

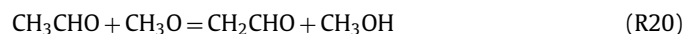
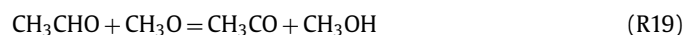
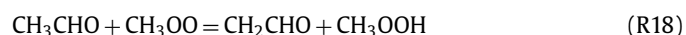
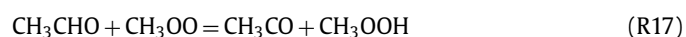
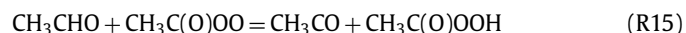
Liquid acetaldehyde (purity 99.5% from Aladdin) was prepared in a stainless-steel vessel cooled by liquid N_2 . After pumping the vessel until there was no N_2 signal detected by the mass spectrometer, pure acetaldehyde vapor was generated by keeping the vessel at a constant temperature of 310 K using heated oil bath. The gas flow rates of acetaldehyde, O_2 and Ar were regulated by mass flow controllers. A three-meter-long stainless-steel pipe was used to connect the mass flow controllers and the reactor to ensure sufficient mixing of inlet gases. The total flow rate of the mixture was 0.5 SLM under all temperature conditions, thus the residence time varied from 3 to 6 seconds. The experimental conditions are summarized in Table 1 for three investigated mixtures with different equivalence ratios (0.5, 1.0 and 4.0).

In this work, the temperature scan was performed at the photon energies of 16.64, 11.7, 11.0 and 10.0 eV. Particularly, at 16.64 eV, cold gas experiments were performed for CO , CO_2 , H_2 and CH_4 . Methyl hydroperoxide and ethane were measured at 10 and 11.7 eV, respectively. Other species were measured at 11 eV. The identification and mole fraction evaluation of intermediates were based on previously reported methods [26]. The photoionization cross sections (PICSs) were referred to our online database [27]. For major products (H_2 , CH_4 , CO , H_2O and CO_2), the uncertainties of measured mole fractions are estimated within $\pm 20\%$. For minor species, the uncertainties was $\pm 50\%$ for species with known PICSs and a factor of 2 for CH_3OOH based on the calculated PICS from Moshhammer et al. [28] and for CH_3OO based on estimation.

3. Theoretical calculations

Rate constants of H-atom abstraction reactions of acetaldehyde by $\text{CH}_3\text{C}(\text{O})\text{OO}$, CH_3OO and CH_3O were calculated theoretically in this work (R15–R20). Note that the order of reactions in the main text is shown in Table S1 of the *Supplementary Materials* which lists the key reactions discussed in this work. R15–R18 can produce $\text{CH}_3\text{C}(\text{O})\text{OOH}$ and CH_3OOH which are degenerate chain-branching triggers under low-temperature conditions

[11,13]. Meanwhile, CH_3O is a key product of the degenerate chain-branching reaction via CH_3OOH decomposition, and its reactions with acetaldehyde (R19 and R20) also play an important role in acetaldehyde consumption [14]. Despite their significance, the aforementioned reactions have not been well characterized in the literature. In recent models [29,30], R16 and R18 were not included, while the rate constants of R15 and R17 were referred to the analogous reaction ($\text{CH}_3\text{CHO} + \text{HO}_2$). But this can be problematic since Carstensen et al. [31,32] concluded in their theoretical investigations that the rate constants decrease in the order of acylperoxy $> \text{HO}_2 >$ alkylperoxy for the H-atom abstraction reactions of alkanes. Furthermore, only limited measurements for the rate constant of R19 below 500 K have been reported [33–35]. Therefore, the rate constants of R15–R20 were theoretically calculated in this work.



3.1. Calculation method

The B2PLYPD3 method [36–38] with cc-pVTZ basis set was used in the geometry optimizations and the calculations of frequencies and external rotational constants for every species with GAUSSIAN 09 software [39]. Single point energy calculations were performed at CCSD(T)/cc-pVTZ level to obtain more accurate energies. The low-frequency modes corresponding to internal rotations were simulated as 1D hindered rotors with hindrance potentials at B2PLYPD3/6-311++g** level. For kinetic predictions, generally, the conventional transition state theory (TST) was applied with asymmetric Eckart tunneling corrections [40]. For cases with relatively lower and flatter barriers, a variational procedure was implemented to ensure that the minimum rate constants were acquired. According to previous theoretical work on the H-atom abstraction reactions of acetaldehyde [41], the pressure-dependent effects can be neglected. High-pressure limit rate constants for R15–R20 were computed using a newly developed kinetic code named MESS [42] over 300–1500 K. The optimized geometries can be found in the *Supplementary Materials*. The uncertainties of computed rate constants of R15–R20 are estimated to be a factor of 2.5 based on the work of Mendes *et al.* using similar calculation method [41].

3.2. Calculated results

Figure 1 shows the calculated PESs for the H-atom abstraction reactions of acetaldehyde by $\text{CH}_3\text{C}(\text{O})\text{OO}$, CH_3OO and CH_3O . All the energies reported here are zero-point corrected electronic energies. At each H site of acetaldehyde, the H-atom abstraction reactions by target radicals begin with the formation of a reactant complex (RC) in the entrance channel via the hydrogen bonding interaction. The RC then proceeds through an H-atom abstraction transition state and forms a product complex (PC) in the exit channel before leading to separate products. The existence of RC and PC was also confirmed on the PESs of $\text{CH}_3\text{CHO} + \text{OH}/\text{HO}_2$ [41]. On each PES, the energy barriers for H-atom abstraction reactions on the CH_3 group are >7 kcal/mol higher than those on the formyl group. The T1 diagnostic values are within 0.028 for all stationary points, implying the unnecessary of multi-reference calculations.

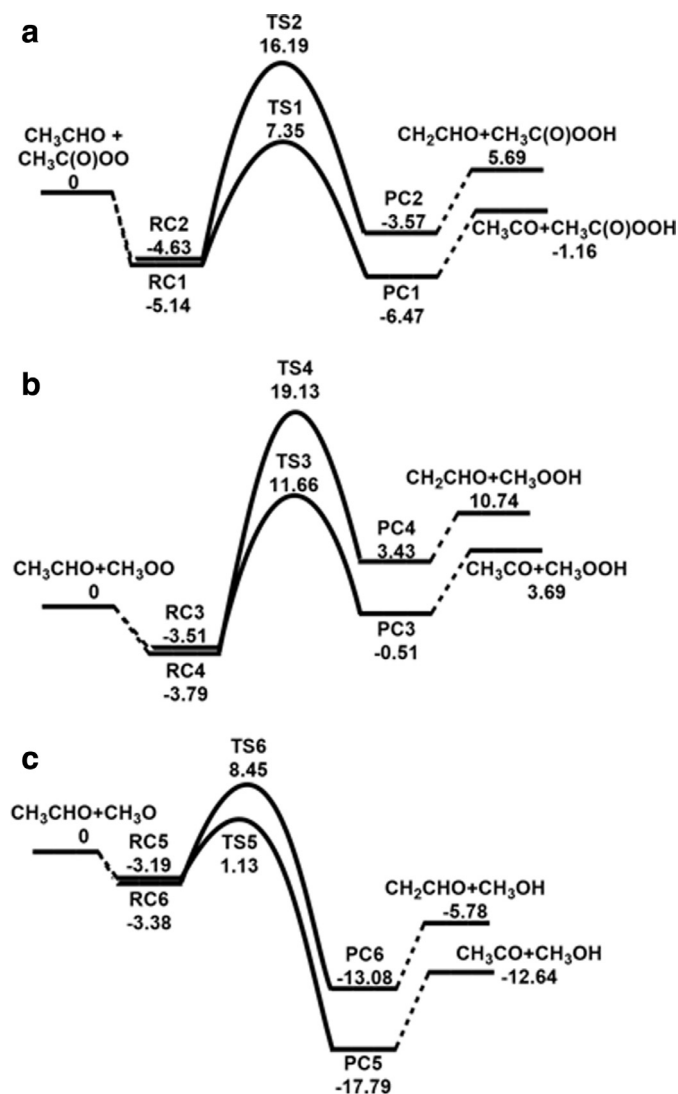


Fig. 1. PESs of the H-atom abstraction reactions of acetaldehyde by (a) $\text{CH}_3\text{C}(\text{O})\text{OO}$, (b) CH_3OO and (c) CH_3O with the CCSD(T)/cc-pVTZ//B2PLYPD3/cc-pVTZ method. The energies are at 0 K in kcal/mol.

The rate constants calculated in this work were compared with the literature values [30,31,33–35,43–46], as shown in Fig. 2. For R15, recent models of Hashemi *et al.* [29] and Metcalfe *et al.* [30] estimated its rate constant by analogy with the same reaction ($\text{CH}_3\text{CHO} + \text{HO}_2 = \text{CH}_3\text{CO} + \text{H}_2\text{O}_2$) from two different sources [43,44]. However, there are large discrepancies between the two rate constants at temperatures below 800 K (see Fig. 2(a)) and the two rate constants from [43,44] are much lower than the measured values of R15 at 300 and 457 K [45,46]. In contrast, the present calculated result is much higher at low temperatures and agrees much better with the measurement of McDowell *et al.* [46] at 300 K. For R17 (see Fig. 2(b)), the recent model [30] also used the rate constant of $\text{CH}_3\text{CHO} + \text{HO}_2$ [43], and the estimated rate constant is higher than the present result. Figure 2(c) presents the comparison between the present calculated result of R19 and those from previous work [33–35]. Our calculated results are in reasonable agreement with the measurements by Fittschen *et al.* [33] in their investigated temperature region, while at 300 K, the present result lies among the measurements of Fittschen *et al.* [33], Kelly *et al.* [34] and Weaver *et al.* [35]. At temperatures higher than 800 K, our value becomes higher than the extrapolation result from Fittschen *et al.* [33]. Furthermore, the present calculations show

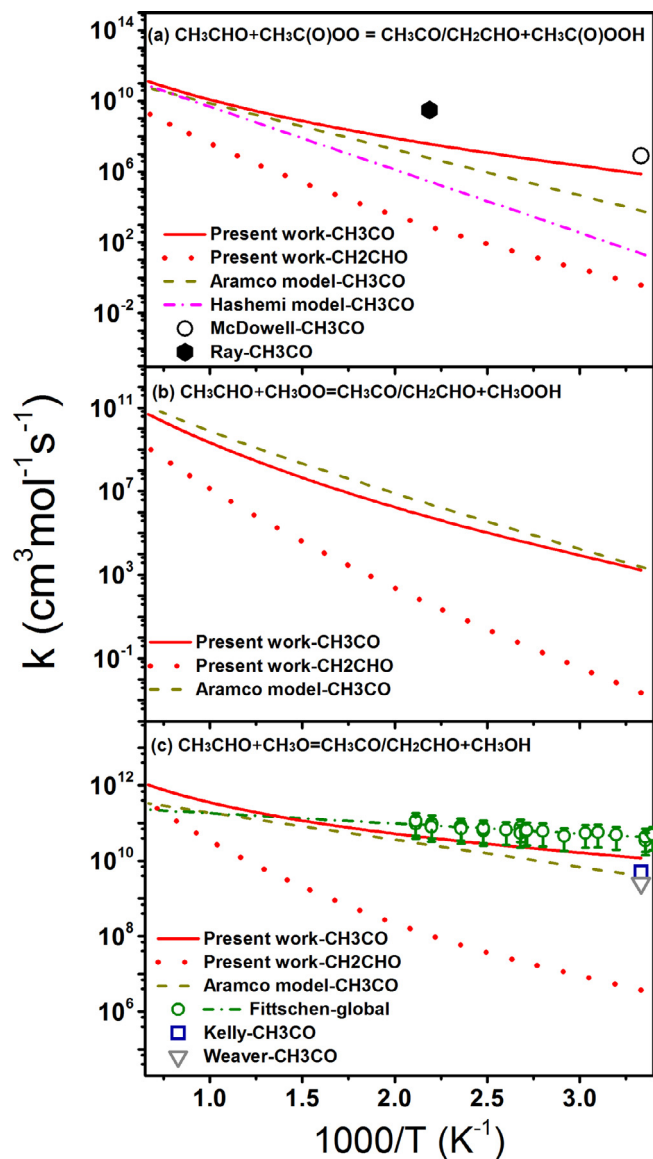


Fig. 2. Plot of the rate constants for H-atom abstraction reactions of acetaldehyde by (a) $\text{CH}_3\text{C}(\text{O})\text{OO}$, (b) CH_3OO and (c) CH_3O . The red solid and dot lines represent the present calculated results for the branching steps of CH_3CO and CH_2CHO , respectively. Other lines represent the estimated results for the branching step of CH_3CO (Aramco model [30] with dash lines and Hashemi model [29] with dash dot line). The symbols represent the experimental results of corresponding reactions in literature: (a) rate constants of R15 from McDowell et al. [46] and Ray et al. [45]; (c) rate constants of R19 from Kelly et al. [34] and Weaver et al. [35] and global rate constant of $\text{CH}_3\text{O} + \text{CH}_3\text{CHO}$ from Fittschen et al. [33].

that below 800 K, the H-atom abstraction reactions of acetaldehyde producing CH_2CHO (R16, R18 and R20) can hardly compete with those producing CH_3CO (R15, R17 and R19). At 800 K, the rate constants of R15, R17 and R19 are 1000, 426 and 25 times larger than those of R16, R18 and R20, respectively. The rate constants calculated in the present work are summarized in Table 2 with modified Arrhenius expression.

4. Kinetic modeling

The present model was constructed based on our previous C1 model [47]. This C1 model was developed based on a well-validated H_2 mechanism [48] and careful evaluations on the elementary reactions involved in $\text{CO}/\text{CH}_2\text{O}/\text{CH}_3\text{OH}$ mechanism. It was validated against comprehensive experimental targets of CH_2O and CH_3OH . The high-temperature mechanism of CH_4 and C2 species was mainly taken from our previous methane model [49]. Besides, since methyl is a key intermediate in the acetaldehyde oxidation, an accurate methyl oxidation mechanism is crucial, especially at low temperatures. In this work, we mainly adopted the low-temperature methyl oxidation mechanism from Hashemi et al. [50]. The thermodynamic data were mainly adopted from Hashemi et al. [29], or calculated using the THERGAS program [51] based on the group and bond additivity methods proposed by Benson [52] for species without reliable literature data. The key reactions incorporated in the sub-mechanism of acetaldehyde are briefly described below, and detailed descriptions on the evaluation of their rate constants can be found in the *Supplementary Materials*.

In the present model, the chain-initiation reactions include unimolecular decomposition reactions and H-atom abstraction reaction of acetaldehyde by O_2 (R1–R7). According to the evaluation in the *Supplementary Materials*, the rate constants of R1–R6 were referred to [6,53] and that of R7 was taken from [43]. As for the chain-propagation reactions, H-atom abstraction reactions (R8–R20) by different radicals, such as H, OH, CH_3 , HO_2 , CH_3O , CH_3OO and $\text{CH}_3\text{C}(\text{O})\text{OO}$ play an important role. Their rate constants were taken from previous studies [41,54,55] and present theoretical calculations.

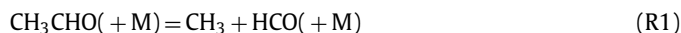


Table 2

Calculated rate constants of R15–R20 in this work over 300–1500 K. Note that the rate constants are in modified Arrhenius expression ($k = A_1 T^{n_1} \exp(-E_1/RT) + A_2 T^{n_2} \exp(-E_2/RT)$) and the fitting errors are within $\pm 5\%$. The units are K, s^{-1} , cm^3 and cal/mol .

Reaction	A_1	n_1	E_1	A_2	n_2	E_2
R15 $\text{CH}_3\text{CHO} + \text{CH}_3\text{C}(\text{O})\text{OO} = \text{CH}_3\text{CO} + \text{CH}_3\text{C}(\text{O})\text{OOH}$	1.92×10^0	3.64	5642	9.81×10^{-4}	4.32	2636.1
R16 $\text{CH}_3\text{CHO} + \text{CH}_3\text{C}(\text{O})\text{OO} = \text{CH}_2\text{CHO} + \text{CH}_3\text{C}(\text{O})\text{OOH}$	2.54×10^{-3}	4.36	1342	3.60×10^{-11}	6.20	7351
R17 $\text{CH}_3\text{CHO} + \text{CH}_3\text{OO} = \text{CH}_3\text{CO} + \text{CH}_3\text{OOH}$	3.22×10^{-1}	3.94	9503	4.99×10^{-6}	4.98	5268.2
R18 $\text{CH}_3\text{CHO} + \text{CH}_3\text{OO} = \text{CH}_2\text{CHO} + \text{CH}_3\text{OOH}$	8.23×10^{-3}	4.28	1681	7.08×10^{-9}	5.66	11,699
R19 $\text{CH}_3\text{CHO} + \text{CH}_3\text{O} = \text{CH}_3\text{CO} + \text{CH}_3\text{OH}$	1.69×10^5	2.04	2353	9.62×10^3	2.50	159
R20 $\text{CH}_3\text{CHO} + \text{CH}_3\text{O} = \text{CH}_2\text{CHO} + \text{CH}_3\text{OH}$	2.65×10^1	3.45	5873	5.64×10^{-6}	4.93	628

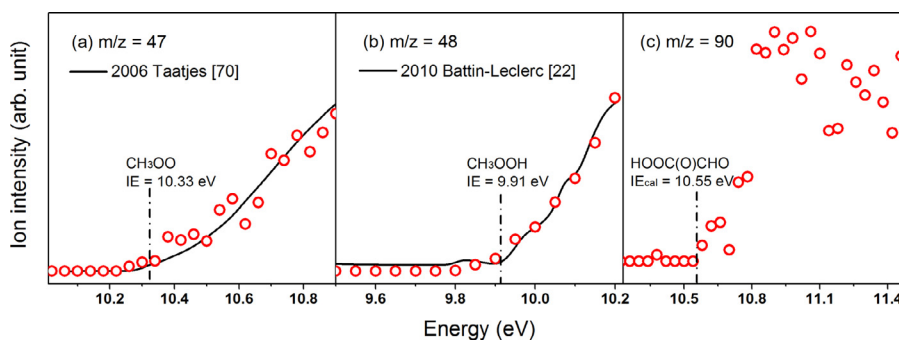
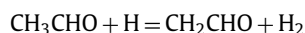
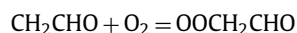


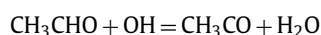
Fig. 3. Measured PIE spectra (symbols) of (a) methylperoxy (CH_3OO), (b) methylhydroperoxide (CH_3OOH) and (c) hydroperoxy glyoxal (HOOC(O)CHO) in this work. Lines represent measurements in other reaction systems or Franck-Condon calculation by previous groups [22,70].



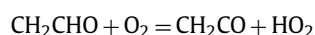
(R9)



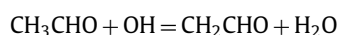
(R32)



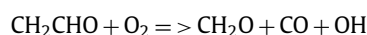
(R10)



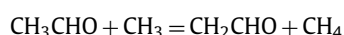
(R33)



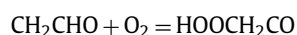
(R11)



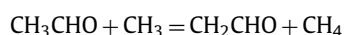
(R34)



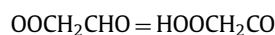
(R12)



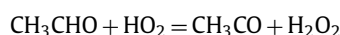
(R35)



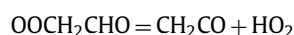
(R13)



(R36)

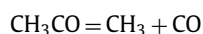


(R14)

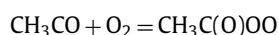


(R37)

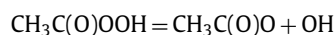
The sub-mechanism of acetyl mainly includes the unimolecular decomposition reaction (R21) and the O_2 addition reactions (R22–R25). Senosiain et al. [4] calculated the PES of CH_3CO and provided pressure-dependent rate constant for R21 which was adopted in the present model. The reaction between CH_3CO and O_2 producing $\text{CH}_3\text{C(O)OO}$ (R22) is an important reaction in atmospheric chemistry since it serves as a step in the conversion of acetaldehyde to peroxyacetylnitrate (a secondary air pollutant) [56]. Lots of studies [56–66] have been carried out for $\text{CH}_3\text{CO} + \text{O}_2$, but many of them were only focused on the kinetics at room temperature. In this model, we adopted the calculation work for R22–R29 from [59].



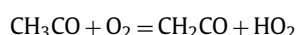
(R21)



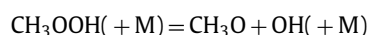
(R22)



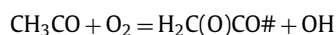
(R40)



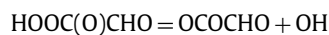
(R23)



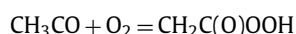
(R41)



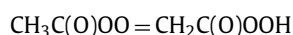
(R24)



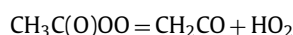
(R42)



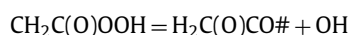
(R25)



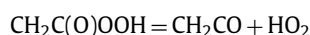
(R26)



(R27)

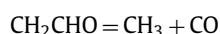


(R28)

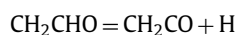


(R29)

The sub-mechanism of CH_2CHO also includes unimolecular decomposition reactions (R30–R31) and O_2 addition reactions (R32–R39). Their rate constants were taken from the calculation work of Senosiain et al. [4] and Lee et al. [65], respectively.



(R30)



(R31)

5. Results and discussion

5.1. Measured and simulated results

In this work, major species (CH_3CHO , O_2 , Ar, CO, H_2O , CH_4 and H_2) and important intermediates, e.g., CH_2O and CH_3OH , were identified. Some reactive intermediates, such as CH_3OO , CH_3OOH and HOOC(O)CHO , were also detected. Figure 3 shows the measured photoionization efficiency (PIE) spectra of CH_3OO , CH_3OOH and HOOC(O)CHO . Barari et al. [15,16] reported the observation of CH_3OO in $\text{CH}_3\text{CHO}/\text{O}_2/\text{Cl}_2/\text{He}$ reaction system with laser photolysis used to generate chlorine atom and form the fuel radicals, but no quantification data was provided for modeling purpose. In this work, the photoionization efficiency (PIE) spectra of CH_3OO agree well with the Franck-Condon calculation of CH_3OO in literature [70], confirming the existence of CH_3OO in the present experiment, as shown in Fig. 3(a). Besides, unambiguous identification of CH_3OOH was achieved based on the measurements of

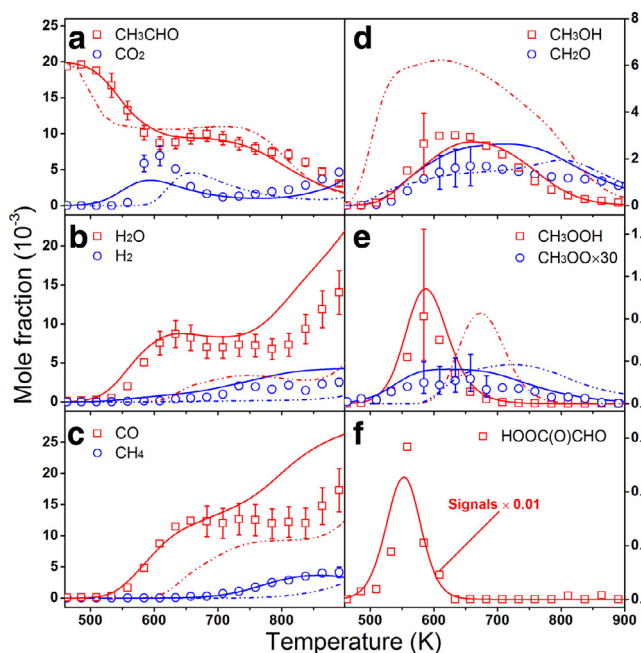


Fig. 4. Measured (symbols) and simulated (lines) mole fraction profiles of oxidation species of acetaldehyde at $\phi = 0.5$ and $P = 710$ Torr. For HOOC(O)CHO , the simulated result is directly compared with the normalized signals. Solid lines: the present model, dash dot lines: POLIMI model [13].

both photoionization mass spectra and PIE spectra. As shown in Fig. 3(b), both the ionization threshold and the signal profile measured in this work agree well with previous CH_3OOH measurement [22], indicating the signal of $m/z = 48$ measured in this work can be attributed to CH_3OOH . Furthermore, Fig. 3(c) shows that the measured ionization threshold of $m/z = 90$ is around 10.54 eV, which is in good agreement with the calculated IE of HOOC(O)CHO (10.55 eV) at the CBS-QB3 level. The calculation method of ionization energy (IE) is as the same as that used by Wang et al. [71]. The observation of HOOC(O)CHO reveals the existence of second O_2 addition process in acetaldehyde oxidation and will be discussed in detail later. The optimized geometries of HOOC(O)CHO and its ion were provided in the *Supplementary Materials*.

The measured and simulated mole fraction profiles of key oxidation species can be found in Figs. 4–6. It is noted that the signals of $m/z = 44$ in the mass spectra at 16.64 eV included both acetaldehyde and CO_2 . The signals of CO_2 were obtained by subtracting the normalized signals of acetaldehyde based on the low photon energy experiments from the total signals of $m/z = 44$, which may introduce additional uncertainties to the results of CO_2 , especially under rich conditions, in which the concentration of CO_2 is relatively low. Recent work of Moshhammer et al. [28] calculated the PICSS of CH_3OOH , which is very helpful for calculating the mole fraction of CH_3OOH . Besides, in order to calculate the mole fraction of CH_3OO , we estimated that its PICSS was equal to CH_3OOH at 11 eV. For the PICSS of HOOC(O)CHO , it is difficult to estimate by analogy. Thus only normalized signals are provided in Fig. 4(f). According to Figs. 4–6, it is concluded that the present model can better capture the temperature windows of acetaldehyde oxidation under all the investigated conditions compared with the recent model [13]. This is mainly because more accurate elementary reaction rate constants involved in the sub-mechanisms of CH_3CO and CH_3 are used in the present model. Furthermore, the present model was also validated against the experimental targets in the literature, including species profiles in shock tube pyrolysis [72–74], JSR oxidation [75] and laminar premixed flat flames [8,76] and global combustion parameters such as laminar burning

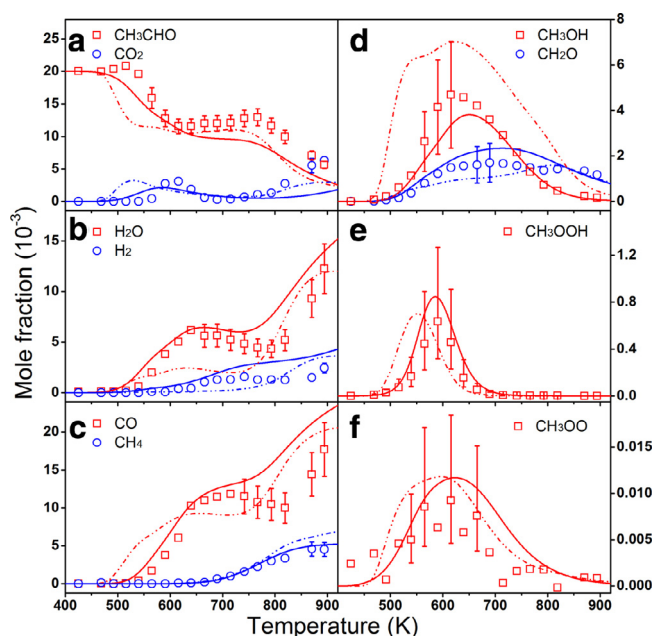


Fig. 5. Measured (symbols) and simulated (lines) mole fraction profiles of oxidation species of acetaldehyde at $\phi = 1.0$ and $P = 720$ Torr. Solid lines: the present model, dash dot dot lines: POLIMI model [13].

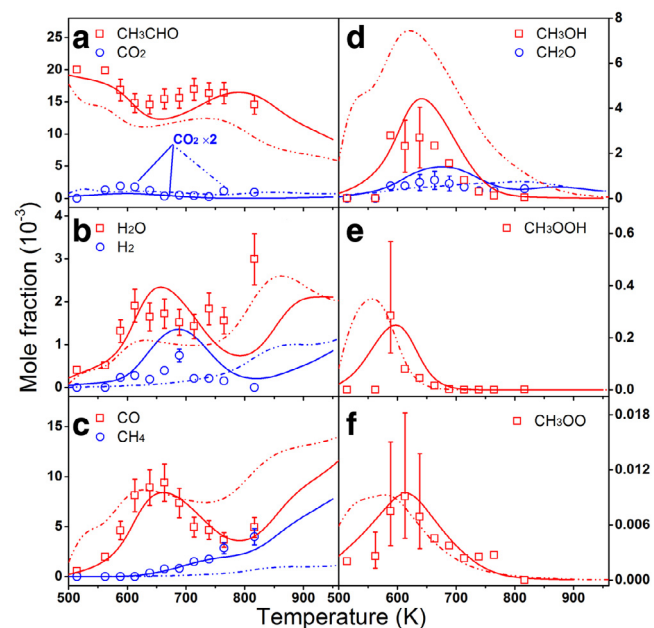


Fig. 6. Measured (symbols) and simulated (lines) mole fraction profiles of oxidation species of acetaldehyde at $\phi = 4.0$ and $P = 720$ Torr. Solid lines: the present model, dash dot dot lines: POLIMI model [13].

velocities [5] and ignition delay times [7,77], which can be found in the *Supplementary Materials*. Generally, the present model has good performance in reproducing all these experimental data.

5.2. Consumption of acetaldehyde and acetyl

According to the rate of production (ROP) analysis, H-atom abstraction reactions of acetaldehyde by different radicals are the most important chain-propagation reactions in acetaldehyde oxidation since they are the dominant pathways for the consumption of acetaldehyde and formation of major products like H_2O , H_2 , CH_4 , etc., as well as the degenerate chain-branching triggers

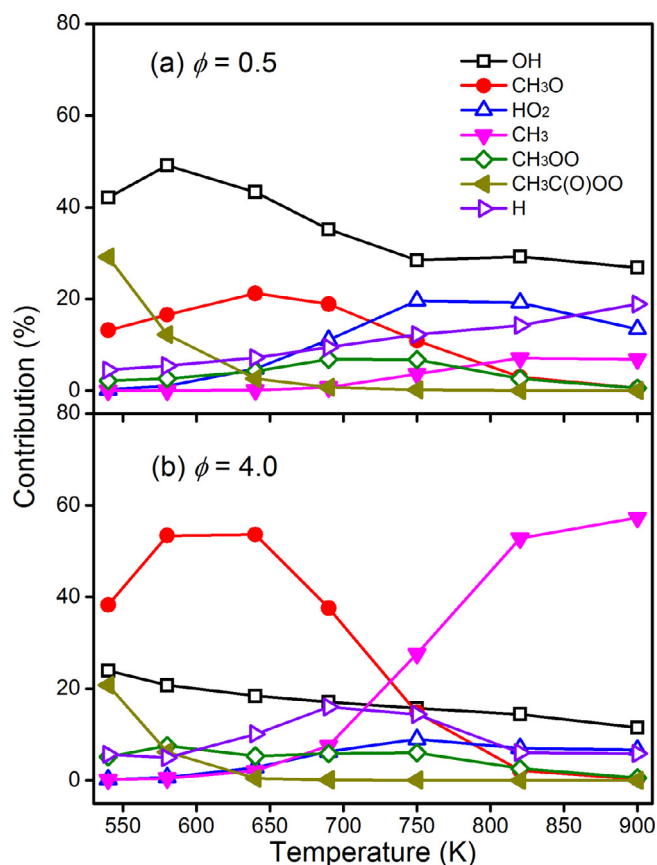


Fig. 7. Contributions of various radicals to the consumption of acetaldehyde at $\phi =$ (a) 0.5 and (b) 4.0.

like $\text{CH}_3\text{C}(\text{O})\text{OOH}$ and CH_3OOH . The H-atom abstraction reactions by various radicals, i.e., OH, CH_3O , HO_2 , CH_3 , CH_3OO , $\text{CH}_3\text{C}(\text{O})\text{OO}$ and H, have different contributions at different temperatures and equivalence ratios. Figure 7 presents the contributions of these radicals to the consumption of acetaldehyde based on the ROP analysis over the temperature range of 540–900 K at equivalence ratios of 0.5 and 4.0.

Under lean conditions, the H-atom abstraction reaction of acetaldehyde by OH is the most important one in the investigated temperature range. Meanwhile the H-atom abstraction reaction by $\text{CH}_3\text{C}(\text{O})\text{OO}$ only contributes largely below 600 K, implying that the H-atom abstraction by $\text{CH}_3\text{C}(\text{O})\text{OO}$ is only important in the low-temperature regime. Besides, in the negative temperature coefficient (NTC) region (around 600–700 K), the H-atom abstraction reaction by CH_3O becomes crucial, consuming nearly 20% of acetaldehyde. However, its important role is replaced by the H-abstraction reactions of HO_2 and H when the temperature increases. In the high-temperature regime, the contributions of H-abstraction reactions by HO_2 and H become important besides that by OH. Each of them contributes nearly 20% to the consumption of acetaldehyde. The H-atom abstraction reactions of acetaldehyde by CH_3 and CH_3OO are less important under the whole investigated temperature region compared with the reactions mentioned above. In contrast, the dominant reactions are quite different under rich conditions. The H-atom abstraction reaction by OH is no longer the most important one. Instead, the H-atom abstraction reaction by CH_3O dominates in the low-temperature and NTC regimes, while the one by CH_3 becomes the most important reaction in the high-temperature regime. Besides, in the low-temperature regime, the H-atom abstraction reaction of acetaldehyde by CH_3OO plays a more important role in the rich than in the lean condition.

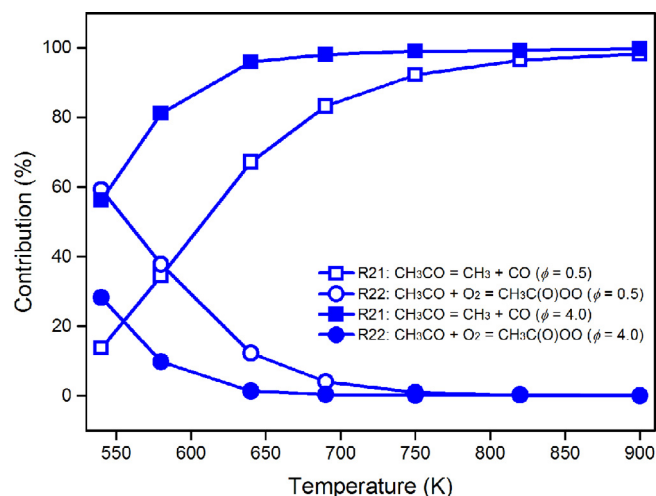


Fig. 8. Contributions of R21 and R22 to the consumption of CH_3CO at $\phi = 0.5$ and 4.0.

The formation of CH_3CO is more favorable than CH_2CHO under the investigated conditions [13,78]. Therefore, the following discussion is focused on the consumption pathways of CH_3CO which can be divided into two classes. One is the unimolecular decomposition reaction which leads to the formation of CH_3 (R21). The other is the O_2 addition reactions which mainly form $\text{CH}_3\text{C}(\text{O})\text{OO}$ through R22. The competition between R21 [4] and R22 [59] depends on both temperature and equivalence ratio, as revealed in Fig. 8. At $\phi = 0.5$, there is a transition temperature around 580 K. Below this temperature, R22 is more favorable which leads to the formation of $\text{CH}_3\text{C}(\text{O})\text{OO}$, indicating the sub-mechanism of $\text{CH}_3\text{C}(\text{O})\text{OO}$ is influential in the low-temperature regime. However, R21 becomes more important at temperatures above 580 K and the formation of CH_3 will be favored. Therefore, the methyl oxidation mechanism will play an important role in the NTC and high-temperature regimes under lean conditions. In contrast, R21 is the dominant consumption pathway of CH_3CO over the whole investigated temperature range under rich condition. Based on the above discussions, the sub-mechanism of $\text{CH}_3\text{C}(\text{O})\text{OO}$ only plays an important role under lean and low-temperature conditions, while the methyl oxidation mechanism is crucial under other investigated conditions in this work.

5.3. Chain-branching pathways under different regimes

To reveal the driving forces in determining the oxidation rate of acetaldehyde, both ROP analysis and sensitivity analysis were performed under different conditions. Four chain-branching pathways were characterized and presented in Fig. 9, including (a) the formation and decomposition of $\text{HOOC}(\text{O})\text{CHO}$, (b) the decomposition of $\text{CH}_3\text{C}(\text{O})\text{OOH}$, (c) the decomposition of CH_3OOH and (d) the decomposition of H_2O_2 . Since the chain-branching agents and pathways are similar under different equivalence ratios in this work, the lean acetaldehyde oxidation will be discussed as an example in this part. Sensitivity analysis was performed at 540 K, 640 K and 750 K at $\phi = 0.5$, corresponding to the low-temperature, NTC and high-temperature regimes, respectively.

The low-temperature regime: At $\phi = 0.5$ and $T < 580$ K, more than 40% CH_3CO can combine O_2 to generate $\text{CH}_3\text{C}(\text{O})\text{OO}$. Subsequently, there are two chain-branching pathways competitive with each other (pathways (a) and (b) in Fig. 9). Pathway (a) is through the isomerization reaction to form QOOH ($\text{CH}_2\text{C}(\text{O})\text{OOH}$), and then QOOH proceeds the second O_2 addition reaction to produce ketohydroperoxide ($\text{HOOC}(\text{O})\text{CHO}$). The observation of

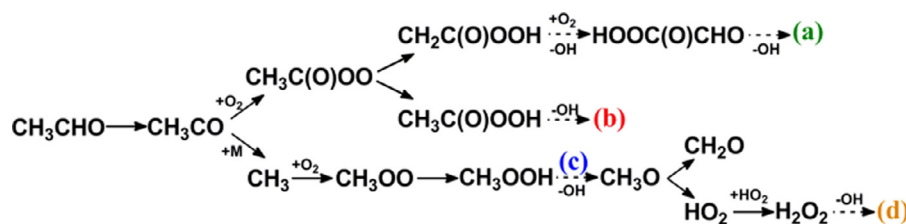


Fig. 9. Reaction networks of the acetaldehyde oxidation. Four chain-branching pathways are listed: (a) formation and decomposition of $\text{HOOC}(\text{O})\text{CHO}$, (b) decomposition of $\text{CH}_3\text{C}(\text{O})\text{OOH}$, (c) decomposition of CH_3OOH and (d) decomposition of H_2O_2 .

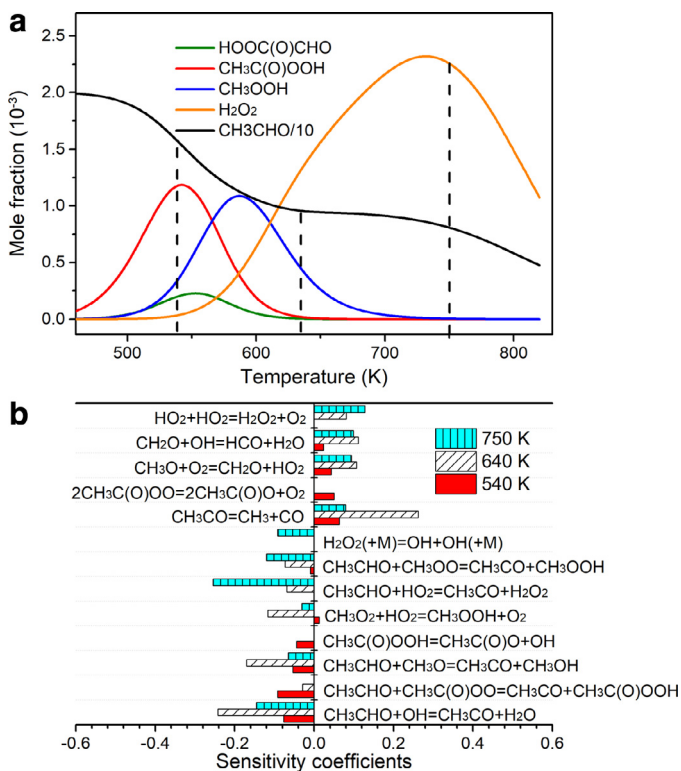


Fig. 10. (a) The predicted mole fractions of chain-branching agents ($\text{HOOC}(\text{O})\text{CHO}$, $\text{CH}_3\text{C}(\text{O})\text{OOH}$, CH_3OOH and H_2O_2), as well as acetaldehyde which is divided by 10 at $\phi = 0.5$ and $P = 710$ Torr; (b) sensitivity analysis of acetaldehyde in the acetaldehyde oxidation at $\phi = 0.5$, $P = 710$ Torr and $T = 540$, 640 and 750 K.

$\text{HOOC}(\text{O})\text{CHO}$ verifies the existence of this pathway in the low-temperature oxidation of acetaldehyde even though acetaldehyde is a much smaller fuel. Pathway (b) is through the bimolecular reaction (R15) to produce $\text{CH}_3\text{C}(\text{O})\text{OOH}$. Subsequently, the decomposition of $\text{CH}_3\text{C}(\text{O})\text{OOH}$, i.e., R40 ($\text{CH}_3\text{C}(\text{O})\text{OOH} = \text{CH}_3\text{C}(\text{O})\text{O} + \text{OH}$), leads to the chain branching. Further consumption of $\text{CH}_3\text{C}(\text{O})\text{O}$ is mainly through unimolecular decomposition reaction, leading to the formation of CH_3 and CO_2 . Figure 10(a) presents the predicted mole fractions of chain-branching agents. As we can see, below 540 K, $\text{CH}_3\text{C}(\text{O})\text{OOH}$ is the dominant chain-branching agent, and its decomposition serves as the main channel for OH formation. In contrast, the concentration of $\text{HOOC}(\text{O})\text{CHO}$ is much less than $\text{CH}_3\text{C}(\text{O})\text{OOH}$, indicating this pathway is not favorable under present conditions. This is within expectation because the carbon chain of acetaldehyde is too short, and only a 5-membered ring can be formed in the transition state from $\text{CH}_3\text{C}(\text{O})\text{OO}$ to $\text{CH}_2\text{C}(\text{O})\text{OOH}$. Figure 10(b) presents the sensitivity analysis of acetaldehyde at $\phi = 0.5$ and different temperatures. As can be seen, R15 ($\text{CH}_3\text{CHO} + \text{CH}_3\text{C}(\text{O})\text{OO} = \text{CH}_3\text{CO} + \text{CH}_3\text{C}(\text{O})\text{OOH}$) is the most sensitive reaction promoting the consumption of acetaldehyde at 540 K since this reaction dominates the formation of $\text{CH}_3\text{C}(\text{O})\text{OOH}$.

At the end of the low-temperature regime, the predominant role of $\text{CH}_3\text{C}(\text{O})\text{OOH}$ for chain-branching is replaced by CH_3OOH gradually, which can be seen in Fig. 10(a).

The NTC region: As temperature rises, the unimolecular decomposition reaction of CH_3CO becomes more important than the O_2 addition reaction. Therefore, the acetaldehyde oxidation is mainly controlled by the low-temperature methyl oxidation mechanism, i.e., pathway (c) in Fig. 9. The decomposition of CH_3OOH serves as the chain-branching reaction. Meanwhile, CH_3O can be formed and it mainly reacts with O_2 to form CH_2O and HO_2 . On the one hand, the subsequent consumption of CH_2O consumes OH. On the other hand, the self-combination of HO_2 forming H_2O_2 leads to the termination of chain reaction. Consequently, the oxidation rate of acetaldehyde is reduced, leading to the formation of NTC region. As we can see in Fig. 10(a), in the NTC region, CH_3OOH is largely consumed while H_2O_2 is gradually accumulated. The sensitivity analysis of acetaldehyde is conducted at 640 K. It is shown that the reactions which lead to the formation of CH_3OOH present large negative sensitivity coefficients towards the consumption of acetaldehyde, such as $\text{CH}_3\text{OO} + \text{HO}_2 = \text{CH}_3\text{OOH} + \text{O}_2$ and $\text{CH}_3\text{CHO} + \text{CH}_3\text{OO} = \text{CH}_3\text{CO} + \text{CH}_3\text{OOH}$, as shown in Fig. 10(b).

The high-temperature regime: As the temperature continues to increase, the acetaldehyde oxidation is still controlled by the methyl oxidation mechanism. Unlike the NTC region, H_2O_2 becomes thermally unstable so that it decomposes to two OH radicals. This reaction serves as a specific chain-branching pathway in the high-temperature regime (pathway (d) in Fig. 9). Figure 10(a) also shows that above 750 K, H_2O_2 begins to decompose and the high-temperature oxidation is triggered. Therefore, the reactions involved in the generation and consumption of H_2O_2 have large sensitivity coefficients to the consumption of acetaldehyde, as shown in Fig. 10(b).

5.4. Methyl oxidation at low and intermediate temperatures

As discussed above, CH_3CO easily decomposes to form CH_3 and CO , especially under elevated temperature and equivalence ratio conditions (see Fig. 8). Therefore, in this part, the kinetics of methyl oxidation is discussed based on the results under rich conditions. The reaction pathways of methyl oxidation are presented in Fig. 11 based on the ROP analysis of acetaldehyde oxidation at $\phi = 4.0$. Among them, the dominant one for methyl oxidation is R43 to form CH_3OO , especially at lower temperatures. Subsequently, there are three possible pathways for the consumption of CH_3OO in the low-temperature and NTC regimes. The most important one is the chain-branching pathway through $\text{CH}_3\text{OO} \rightarrow \text{CH}_3\text{OOH} \rightarrow \text{CH}_3\text{O}$. The other two reaction pathways are competitive with it. They combine CH_3 or CH_3OO to form two CH_3O radicals. It is noticed that all the three pathways transfer CH_3OO to CH_3O . CH_3O also has various consumption pathways. It can serve as an abstractor to react with CH_2O or acetaldehyde. In this way, CH_3OH will be generated. Alternatively, it reacts with O_2 or decomposes to form CH_2O . In this work, CH_3OO , CH_3OOH , CH_3OH and CH_2O were measured, as high-

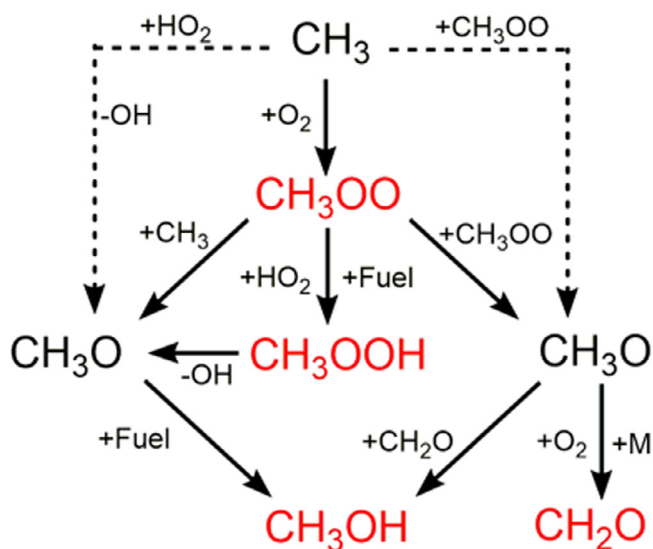


Fig. 11. Reaction network of methyl oxidation in the low-temperature and NTC regimes of the rich acetaldehyde oxidation. The species measured in this work are highlighted. Minor pathways under this temperature region are shown as dashed arrows.

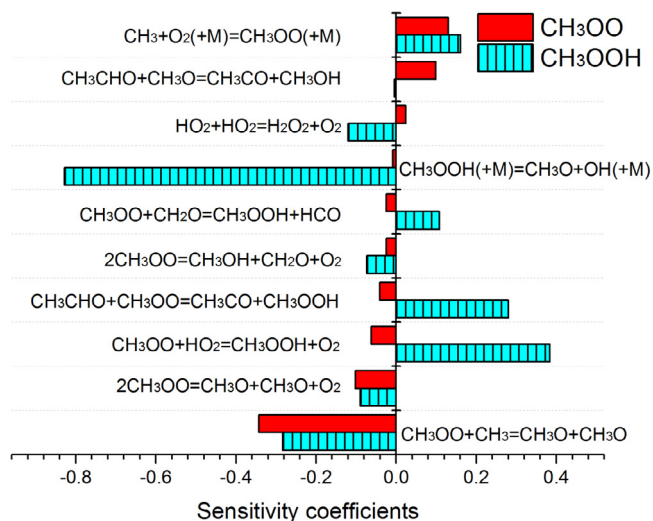


Fig. 12. Sensitivity analysis of CH_3OO and CH_3OOH in the rich acetaldehyde oxidation at 640 K.

lighted in Fig. 11, which provides stringent constraints for the validation of present model. Figure 12 presents the sensitivity analysis of CH_3OO and CH_3OOH in the acetaldehyde oxidation at $\phi = 4.0$ and 640 K. This temperature is within the NTC region, where the methyl oxidation mechanism plays an important role. As can be seen, the reactions involved in the formation and consumption of CH_3OOH have large sensitivity coefficients, e.g., R44, R17 ($\text{CH}_3\text{CHO} + \text{CH}_3\text{OO} = \text{CH}_3\text{CO} + \text{CH}_3\text{OOH}$) and R41 ($\text{CH}_3\text{OOH} (+\text{M}) = \text{CH}_3\text{O} + \text{OH} (+\text{M})$). Besides, R43 and R45, which are involved in the formation and consumption of CH_3OO , also have large sensitivity coefficients for both CH_3OOH and CH_3OO . Therefore, this work can also provide experimental data for the validation of methyl oxidation mechanism under very low-temperature conditions.



6. Conclusions

The JSR oxidation of acetaldehyde was investigated at temperatures from 460 to 900 K, equivalence ratios from 0.5 to 4.0 and pressures of 710–720 Torr. The identification and quantification of the intermediates were obtained using SVUV-PIMS. Besides major products, reactive species like CH_3OO , CH_3OOH , and $\text{HOOC}(\text{O})\text{CHO}$ were measured in this work, providing new insights into the oxidation mechanism of acetaldehyde at low and intermediate temperatures. The H-atom abstraction reactions of acetaldehyde by $\text{CH}_3\text{C}(\text{O})\text{OO}$, CH_3OO and CH_3O were calculated using B2PLYPD3/cc-pVTZ method, with the single point energy calculations performed at CCSD(T)/cc-pVTZ level. Rate constants of these reactions were obtained based on the variational TST method with asymmetric Eckart tunneling corrections. A kinetic model of acetaldehyde was developed based on the present experimental measurements, theoretical calculations and fundamental studies of CH_3CHO in literature. The present model could reproduce the temperature windows of acetaldehyde consumption and the formation of those intermediates measured in this work. The main chain-branching pathways in the acetaldehyde oxidation are the decomposition of $\text{CH}_3\text{C}(\text{O})\text{OOH}$ and CH_3OOH , while the second O_2 addition reaction and its subsequent pathways can hardly influence the predictions under present investigated conditions. Besides, the sub-mechanism of $\text{CH}_3\text{C}(\text{O})\text{OO}$ is only crucial in the low-temperature regime, while the methyl oxidation mechanism dominants in the NTC region regardless of the equivalence ratios. The reactions related to methyl oxidation are very sensitive to CH_3OO and CH_3OOH under present experimental conditions, especially in the NTC region. Therefore the present data provide additional validation targets for the low-temperature methyl oxidation mechanism.

Acknowledgments

The authors are grateful for the funding support from National Natural Science Foundation of China (91541201, 51622605, 11675111) and Science and Technology Commission of Shanghai Municipality (No. 17XD1402000). The authors appreciate Dr. Robin Shannon and Dr. David R. Glowacki for providing the complete calculated results of $\text{CH}_3\text{CO} + \text{O}_2$ system, Dr. Zhandong Wang for discussions on the experiments and Mr. Tianyu Li, Ms. Wei Li and Mr. Jiaobiao Zou for technical assistance.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2018.01.027](https://doi.org/10.1016/j.combustflame.2018.01.027).

References

- [1] K.R. Smith, H. Frumkin, K. Balakrishnan, C.D. Butler, Z.A. Chafe, I. Fairlie, P. Kinney, T. Kjellstrom, D.L. Mauzerall, T.E. McKone, A.J. McMichael, M. Schneider, Energy and human health, *Annu. Rev. Public Health* 34 (2013) 159–188.
- [2] G. Fontaras, G. Karavalakis, M. Kousoulidou, L. Ntziachristos, E. Bakeas, S. Stournas, Z. Samaras, Effects of low concentration biodiesel blends application on modern passenger cars. Part 2: Impact on carbonyl compound emissions, *Environ. Pollut.* 158 (2010) 2496–2503.
- [3] L.L.N. Guarieiro, A.F. de Souza, E.A. Torres, J.B. de Andrade, Emission profile of 18 carbonyl compounds, CO , CO_2 , and NO_x emitted by a diesel engine fuelled with diesel and ternary blends containing diesel, ethanol and biodiesel or vegetable oils, *Atmos. Environ.* 43 (2009) 2754–2761.
- [4] J.P. Senosiain, S.J. Klippenstein, J.A. Miller, Pathways and rate coefficients for the decomposition of vinoxy and acetyl radicals, *J. Phys. Chem. A* 110 (2006) 5772–5781.
- [5] M. Christensen, M.T. Abebe, E.J.K. Nilsson, A.A. Konnov, Kinetics of premixed acetaldehyde + air flames, *Proc. Combust. Inst.* 35 (2015) 499–506.

- [6] R. Sivaramakrishnan, J.V. Michael, L.B. Harding, S.J. Klippenstein, Resolving some paradoxes in the thermal decomposition mechanism of acetaldehyde, *J. Phys. Chem. A* 119 (2015) 7724–7733.
- [7] R. Mével, K. Chatelain, L. Catoire, W. Green, J. Shepherd, Chemical kinetics of acetaldehyde pyrolysis and oxidation, 9th US National Combustion Meeting (2015).
- [8] T. Tao, W.Y. Sun, B. Yang, N. Hansen, K. Moshhammer, C.K. Law, Investigation of the chemical structures of laminar premixed flames fueled by acetaldehyde, *Proc. Combust. Inst.* 36 (2017) 1287–1294.
- [9] M. Christensen, A.A. Konnov, Laminar burning velocity of diacetyl + air flames. Further assessment of combustion chemistry of ketene, *Combust. Flame* 178 (2017) 97–110.
- [10] W.H. Perkin, LVIII. Some observations on the luminous incomplete combustion of ether and other organic bodies, *J. Chem. Soc., Trans* 41 (1882) 363–367.
- [11] M. Pelucchi, A. El Ziani, M. Mensi, E. Ranzi, A. Frassoldati, T. Faravelli, Kinetic modeling of the low temperature cool flames of acetaldehyde in a well stirred reactor, Meeting of the Italian Section of the Combustion Institute (2015).
- [12] P. Gray, J.F. Griffiths, S.M. Hasko, P.G. Lignola, Oscillatory Ignitions and cool flames accompanying the non-isothermal oxidation of acetaldehyde in a well stirred flow reactor, *Proc. R. Soc. Lond., A* 374 (1981) 313.
- [13] M. Pelucchi, E. Ranzi, A. Frassoldati, T. Faravelli, Alkyl radicals rule the low temperature oxidation of long chain aldehydes, *Proc. Combust. Inst.* (2017).
- [14] E.W. Kaiser, C.K. Westbrook, W.J. Pitz, Acetaldehyde oxidation in the negative temperature coefficient regime: Experimental and modeling results, *Int. J. Chem. Kinet.* 18 (1986) 655–688.
- [15] G. Barari, B. Koroglu, A.E. Masunov, S. Vasu, ASME, Combustion of aldehydes in the negative temperature coefficient region: products and pathways, ASME Turbo Expo: Turbine Technical Conference and Exposition, 3 (2016), p. 2016.
- [16] G. Barari, B. Koroglu, A.E. Masunov, S. Vasu, Products and pathways of aldehydes oxidation in the negative temperature coefficient region, *J. Energ. Res. Technol.* 139 (2016) 012203–012203-9.
- [17] M.P. Halstead, A. Prothero, C.P. Quinn, A mathematical model of the cool-flame oxidation of acetaldehyde, *Proc. R. Soc. London, A* 322 (1971) 377.
- [18] P.G. Felton, B.F. Gray, N. Shank, Low temperature oxidation in a stirred flow reactor—II. Acetaldehyde (theory), *Combust. Flame* 27 (1976) 363–376.
- [19] J. Cavanagh, R.A. Cox, G. Olson, Computer modeling of cool flames and ignition of acetaldehyde, *Combust. Flame* 82 (1990) 15–39.
- [20] C. Gibson, P. Gray, J.F. Griffiths, S.M. Hasko, Spontaneous ignition of hydrocarbon and related fuels: A fundamental study of thermokinetic interactions, *Proc. Combust. Inst.* 20 (1985) 101–109.
- [21] Z.Y. Zhou, X.W. Du, J.Z. Yang, Y.Z. Wang, C.Y. Li, S. Wei, L.L. Du, Y.Y. Li, F. Qi, Q.P. Wang, The vacuum ultraviolet beamline/endstations at NSRL dedicated to combustion research, *J. Synchrotron Rad.* 23 (2016) 1035–1045.
- [22] F. Battin-Leclerc, O. Herbinet, P.-A. Glaude, R. Fournet, Z.Y. Zhou, L.L. Deng, H.J. Guo, M.F. Xie, F. Qi, Experimental confirmation of the low-temperature oxidation scheme of alkanes, *Angew. Chem. Int. Ed.* 49 (2010) 3169–3172.
- [23] D. Matras, J. Villiermaux, Un réacteur continu parfaitement agité par jets gazeux pour l'étude cinétique de réactions chimiques rapides, *Chem. Eng. Sci.* 28 (1973) 129–137.
- [24] R. Porter, P.-A. Glaude, F. Buda, F. Battin-Leclerc, A tentative modeling study of the effect of wall reactions on oxidation phenomena, *Energy Fuels* 22 (2008) 3736–3743.
- [25] M. Chambon, P.M. Marquaire, G.M. Côme, The formation of hydrocarbons in the high temperature reaction of chlorine-methane mixtures, *C1 Molecule Chem.* 2 (1987) 47–59.
- [26] O. Herbinet, F. Battin-Leclerc, S. Bax, H.L. Gall, P.-A. Glaude, R. Fournet, Z.Y. Zhou, L.L. Deng, H.J. Guo, M.F. Xie, F. Qi, Detailed product analysis during the low temperature oxidation of *n*-butane, *Phys. Chem. Chem. Phys.* 13 (2011) 296–308.
- [27] Photonization Cross Section Database (Version 2.0), National Synchrotron Radiation Laboratory, Hefei, China, 2017, <http://flame.nsrl.ustc.edu.cn/en/database.htm>.
- [28] K. Moshhammer, A.W. Jasper, D.M. Popolan-Vaida, Z.D. Wang, V.S. Bhavani Shankar, L. Ruwe, C.A. Taatjes, P. Dagaut, N. Hansen, Quantification of the keto-hydroperoxide (HOOC_2OCHO) and other elusive intermediates during low-temperature oxidation of dimethyl ether, *J. Phys. Chem. A* 120 (2016) 7890–7901.
- [29] H. Hashemi, J.G. Jacobsen, C.T. Rasmussen, J.M. Christensen, P. Glarborg, S. Gersen, M. van Essen, H.B. Levinsky, S.J. Klippenstein, High-pressure oxidation of ethane, *Combust. Flame* 182 (2017) 150–166.
- [30] W.K. Metcalfe, S.M. Burke, S.S. Ahmed, H.J. Curran, A hierarchical and comparative kinetic modeling study of C_1 – C_2 hydrocarbon and oxygenated fuels, *Int. J. Chem. Kinet.* 45 (2013) 638–675.
- [31] H.-H. Carstensen, A.M. Dean, O. Deutschmann, Rate constants for the H abstraction from alkanes (R-H) by radicals: a systematic study on the impact of R and R', *Proc. Combust. Inst.* 31 (2007) 149–157.
- [32] H.H. Carstensen, A.M. Dean, Rate constants for the abstraction reactions $\text{RO}_2 + \text{C}_2\text{H}_6$; $\text{R} = \text{H}, \text{CH}_3$, and C_2H_5 , *Proc. Combust. Inst.* 30 (2005) 995–1003.
- [33] C. Fittschen, B. Delcroix, N. Gomez, P. Devolder, Rate constants for the reactions of CH_3O with CH_2O , CH_3CHO and *i*- C_4H_{10} , *J. Chim. Phys. Pcb* 95 (1998) 2129–2142.
- [34] N. Kelly, J. Hecklen, Rate coefficient for the reaction of CH_3O with CH_3CHO at 25°C, *J. Photochem.* 8 (1978) 83–90.
- [35] J. Weaver, J. Meagher, R. Shortridge, J. Hecklen, The oxidation of acetyl radicals, *J. Photochem.* 4 (1975) 341–360.
- [36] L. Goerigk, S. Grimme, A thorough benchmark of density functional methods for general main group thermochemistry, kinetics, and noncovalent interactions, *Phys. Chem. Chem. Phys.* 13 (2011) 6670–6688.
- [37] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H–Pu, *J. Chem. Phys.* 132 (2010) 154104.
- [38] S. Grimme, Semiempirical hybrid density functional with perturbative second-order correlation, *J. Chem. Phys.* 124 (2006) 034108.
- [39] G.W.T.M.J. Frisch, H.B. Schlegel, G.E. Scuseria, J.R.C.M.A. Robb, G. Scalmani, V. Barone, B. Mennucci, H.N.G.A. Petersson, M. Caricato, X. Li, H.P. Hratchian, J.B.A.F. Izmaylov, G. Zheng, J.L. Sonnenberg, M. Hada, K.T.M. Ehara, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, O.K.Y. Honda, H. Nakai, T. Vreven, J.A. Montgomery, Jr., F.O.J.E. Peralta, M. Bearpark, J.J. Heyd, E. Brothers, V.N.S.K.N. Kudin, T. Keith, R. Kobayashi, J. Normand, A.R.K. Raghavachari, J.C. Burant, S.S. Iyengar, J. Tomasi, N.R.M. Cossi, J.M. Millam, M. Klene, J.E. Knox, J.B. Cross, C.A.V. Bakken, J. Jaramillo, R. Gomperts, R.E. Stratmann, A.J.A.O. Yazyev, R. Cammi, C. Pomelli, J.W. Ochterski, K.M.R.L. Martin, V.G. Zakrzewski, G.A. Voth, J.J.D.P. Salvador, S. Dapprich, A.D. Daniels, J.B.F.O. Farkas, J.V. Ortiz, J. Cioslowski, D.J. Fox, Gaussian 09, Revision D.01, Gaussian, Inc., Wallingford CT, 2013.
- [40] C. Eckart, The penetration of a potential barrier by electrons, *Phys. Rev.* 35 (1930) 1303–1309.
- [41] J. Mendes, C.-W. Zhou, H.J. Curran, Theoretical chemical kinetic study of the H-atom abstraction reactions from aldehydes and acids by H atoms and OH , HO_2 , and CH_3 radicals, *J. Phys. Chem. A* 118 (2014) 12089–12104.
- [42] Y. Georgievskii, J.A. Miller, M.P. Burke, S.J. Klippenstein, Reformulation and solution of the master equation for multiple-well chemical reactions, *J. Phys. Chem. A* 117 (2013) 12146–12154.
- [43] D.L. Baulch, C.T. Bowman, C.J. Cobos, R.A. Cox, T. Just, J.A. Kerr, M.J. Pilling, D. Stocker, J. Troe, W. Tsang, R.W. Walker, J. Warnatz, Evaluated kinetic data for combustion modeling: Supplement II, *J. Phys. Chem. Ref. Data* 34 (2005) 757–1397.
- [44] M. Altarawneh, A.H. Al-Muhtaseb, B.Z. Dlugogorski, E.M. Kennedy, J.C. Mackie, Rate constants for hydrogen abstraction reactions by the hydroperoxyl radical from methanol, ethanol, acetaldehyde, toluene, and phenol, *J. Comput. Chem.* 32 (2011) 1725–1733.
- [45] D.J.M. Ray, D.J. Waddington, Epoxidation of alkenes in the gas phase, *J. Phys. Chem.* 76 (1972) 3319–3320.
- [46] C.A. McDowell, L.K. Sharples, The photochemical oxidation of aldehydes in the gaseous phase: part III. The absolute values of the velocity constants for the propagating and terminating steps in the photochemical oxidation of acetaldehyde and propionaldehyde, *Can. J. Chem.* 36 (1958) 268–278.
- [47] X.Y. Zhang, G.Q. Wang, J.B. Zou, Y.Y. Li, W. Li, T.Y. Li, H.F. Jin, Z.Y. Zhou, Y.-Y. Lee, Investigation on the oxidation chemistry of methanol in laminar premixed flames, *Combust. Flame* 180 (2017) 20–31.
- [48] H. Hashemi, J.M. Christensen, S. Gersen, P. Glarborg, Hydrogen oxidation at high pressure and intermediate temperatures: Experiments and kinetic modeling, *Proc. Combust. Inst.* 35 (2015) 553–560.
- [49] H. Jin, A. Frassoldati, Y. Wang, X. Zhang, M. Zeng, Y. Li, F. Qi, A. Cuoci, T. Faravelli, Kinetic modeling study of benzene and PAH formation in laminar methane flames, *Combust. Flame* 162 (2015) 1692–1711.
- [50] H. Hashemi, J.M. Christensen, S. Gersen, H. Levinsky, S.J. Klippenstein, P. Glarborg, High-pressure oxidation of methane, *Combust. Flame* 172 (2016) 349–364.
- [51] C. Muller, V. Michel, G. Scacchi, G. Côme, Estimation of thermodynamic data, *J. Chim. Phys.* 92 (1995) 1154–1178.
- [52] S.W. Benson, Thermochemical Kinetics, Wiley, New York, 1976, pp. 19–78.
- [53] L.B. Harding, Y. Georgievskii, S.J. Klippenstein, Roaming radical kinetics in the decomposition of acetaldehyde, *J. Phys. Chem. A* 114 (2010) 765–777.
- [54] R. Sivaramakrishnan, J.V. Michael, S.J. Klippenstein, Direct observation of roaming radicals in the thermal decomposition of acetaldehyde, *J. Phys. Chem. A* 114 (2010) 755–764.
- [55] P.H. Taylor, T. Yamada, P. Marshall, The reaction of OH with acetaldehyde and deuterated acetaldehyde: Further insight into the reaction mechanism at both low and elevated temperatures, *Int. J. Chem. Kinet.* 38 (2006) 489–495.
- [56] C.B.M. Gross, T.J. Dillon, J.N. Crowley, Pressure dependent OH yields in the reactions of CH_3CO and HOCH_2CO with O_2 , *Phys. Chem. Chem. Phys.* 16 (2014) 10990–10998.
- [57] V.C. Papadimitriou, E.S. Karafas, T. Gierczak, J.B. Burkholder, $\text{CH}_3\text{CO} + \text{O}_2 + \text{M}$ ($\text{M} = \text{He}, \text{N}_2$) reaction rate coefficient measurements and implications for the OH radical product yield, *J. Phys. Chem. A* 119 (2015) 7481–7497.
- [58] J.W. Allen, C.F. Goldsmith, W.H. Green, Automatic estimation of pressure-dependent rate coefficients, *Phys. Chem. Chem. Phys.* 14 (2012) 1131–1155.
- [59] S.A. Carr, D.R. Glowacki, C.H. Liang, M.T. Baeza-Romero, M.A. Blitz, M.J. Pilling, P.W. Seakins, Experimental and modeling studies of the pressure and temperature dependences of the kinetics and the OH yields in the acetyl + O_2 reaction, *J. Phys. Chem. A* 115 (2011) 1069–1085.
- [60] A. Maranzana, J.R. Barker, G. Tonachini, Master equation simulations of competing unimolecular and bimolecular reactions: application to OH production in the reaction of acetyl radical with O_2 , *Phys. Chem. Chem. Phys.* 9 (2007) 4129–4141.
- [61] G. Kovacs, J. Zador, E. Farkas, R. Nadasdi, I. Szilagyi, S. Dobe, T. Berces, F. Marta, G.Y. Lendvay, Kinetics and mechanism of the reactions of CH_3CO and $\text{CH}_3\text{C}(\text{O})\text{CH}_2$ radicals with O_2 . Low-pressure discharge flow experiments and quantum chemical computations, *Phys. Chem. Chem. Phys.* 9 (2007) 4142–4154.

- [62] S.A. Carr, M.T. Baeza-Romero, M.A. Blitz, M.J. Pilling, D.E. Heard, P.W. Seakins, OH yields from the $\text{CH}_3\text{CO}+\text{O}_2$ reaction using an internal standard, *Chem. Phys. Lett.* 445 (2007) 108–112.
- [63] P. Devolder, S. Dusanter, B. Lemoine, C. Fittschen, About the co-product of the OH radical in the reaction of acetyl with O_2 below atmospheric pressure, *Chem. Phys. Lett.* 417 (2006) 154–158.
- [64] H. Hou, A. Li, H.Y. Hu, Y. Li, H. Li, B.S. Wang, Mechanistic and kinetic study of the $\text{CH}_3\text{CO}+\text{O}_2$ reaction, *J. Chem. Phys.* (2005) 122.
- [65] J. Lee, J.W. Bozzelli, Thermochemical and kinetic analysis of the formyl methyl radical plus O_2 reaction system, *J. Phys. Chem. A* 107 (2003) 3778–3791.
- [66] J.W. Lee, C.J. Chen, J.W. Bozzelli, Thermochemical and kinetic analysis of the acetyl radical $\text{CH}_3\text{CO}+\text{O}_2$ reaction system, *J. Phys. Chem. A* 106 (2002) 7155–7170.
- [67] K.A. Sahetchian, R. Rigny, J. Tardieu de Maleissye, L. Batt, M. Anwar Khan, S. Mathews, The pyrolysis of organic hydroperoxides (ROOH), *Proc. Combust. Inst.* 24 (1992) 637–643.
- [68] A.W. Jasper, S.J. Klippenstein, L.B. Harding, Theoretical rate coefficients for the reaction of methyl radical with hydroperoxyl radical and for methylhydroperoxide decomposition, *Proc. Combust. Inst.* 32 (2009) 279–286.
- [69] CHEMKIN-PRO 15092, Reaction Design: San Diego, 2009.
- [70] G. Meloni, P. Zou, S.J. Klippenstein, M. Ahmed, S.R. Leone, C.A. Taatjes, D.L. Osborn, Energy-resolved photoionization of alkylperoxy radicals and the stability of their cations, *J. Am. Chem. Soc.* 128 (2006) 13559–13567.
- [71] Z. Wang, L. Zhang, K. Moshhammer, D.M. Popolan-Vaida, V.S.B. Shankar, A. Lucassen, C. Hemken, C.A. Taatjes, S.R. Leone, K. Kohse-Hoeinghaus, N. Hansen, P. Dagaut, S.M. Sarathy, Additional chain-branching pathways in the low-temperature oxidation of branched alkanes, *Combust. Flame* 164 (2016) 386–396.
- [72] S.K. Wang, D.F. Davidson, R.K. Hanson, High-temperature laser absorption diagnostics for CH_2O and CH_3CHO and their application to shock tube kinetic studies, *Combust. Flame* 160 (2013) 1930–1938.
- [73] T. Bentz, F. Striebel, M. Olzmann, Shock-tube study of the thermal decomposition of CH_3CHO and $\text{CH}_3\text{CHO}+\text{H}$ reaction, *J. Phys. Chem. A* 112 (2008) 6120–6124.
- [74] Y. Hidaka, S. Kubo, T. Hoshikawa, H. Wakamatsu, Shock-tube study of acetaldehyde pyrolysis, *Shock Waves* (2005) 603–608.
- [75] P. Dagaut, M. Reuillon, D. Voisin, M. Cathonnet, M. McGuinness, J.M. Simmie, Acetaldehyde oxidation in a JSR and ignition in shock waves: experimental and comprehensive kinetic modeling, *Combust. Sci. Technol.* 107 (1995) 301–316.
- [76] N. Leplat, J. Vandooren, Experimental investigation and numerical simulation of the structure of $\text{CH}_3\text{CHO}/\text{O}_2/\text{Ar}$ flames at different equivalence ratios, *Combust. Sci. Technol.* 182 (2010) 436–448.
- [77] K. Yasunaga, S. Kubo, H. Hoshikawa, T. Kamesawa, Y. Hidaka, Shock-tube and modeling study of acetaldehyde pyrolysis and oxidation, *Int. J. Chem. Kinet.* 40 (2008) 73–102.
- [78] G. da Silva, J.W. Bozzelli, Enthalpies of formation, bond dissociation energies, and molecular structures of the n-aldehydes (acetaldehyde, propanal, butanal, pentanal, hexanal, and heptanal) and their radicals, *J. Phys. Chem. A* 110 (2006) 13058–13067.